

OCR Further Mathematics
Pure Core - Differential Equations
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



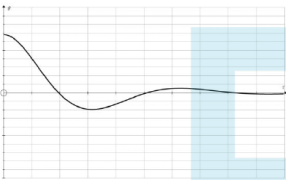
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To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.



Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a	i	For SHM $\lambda = 0$	B1 (AO 3.3) [1]		
		ii	The door should close, but in SHM the motion continues indefinitely	B1 (AO 3.5b) [1]		
	b		Over- or critical- damping implies $\lambda^2 - 12 \geq 0$ So $\lambda \geq 2\sqrt{3}$	M1 (AO 3.3) A1 (AO 3.4) [2]	Consider discriminant with $\geq$ or $>$ Ignore $\lambda \leq -2\sqrt{3}$	
	c		e.  g.	B1 (AO 3.4) [1]	Graph of under-damped system. Start anywhere non-zero on θ -axis with zero gradient. Each peak must be lower than before At least two peaks (not including start point) The graph must look as though it is approaching the t axis	
			Total	5		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	$F = ma = 2 \frac{dv}{dt} = 4e^{-2t} - kv$ $t = \ln 2, v = 0.5, F = 0 \Rightarrow 0 = 1 - 0.5k$ $k = 2 \Rightarrow 2 \frac{dv}{dt} = 4e^{-2t} - 2v \Rightarrow \frac{dv}{dt} + v = 2e^{-2t}$	M1 (AO3.3) M1 (AO2.2a) A1 (AO1.1) [3]	Use of NII with m and a replaced and with 2 forces, the given force and kv Use of given conditions to derive an equation in k AG	$F=ma$ can be implicit here Can be done first Complete argument including $F=ma$	
	b	$IF = e^{\int 1 dt} = e^t$ $e^t \frac{dv}{dt} + e^t v = \frac{d}{dt}(e^t v) = e^t \times 2e^{-2t}$ $e^t v = \int 2e^{-t} dt = -2e^{-t} + c$ $t = 0, v = 0 \Rightarrow c = 2$ $v = 2e^{-t} - 2e^{-2t}$	*B1 (AO1.1) *M1 (AO1.1) A1 (AO1.1) dep*M1 (AO3.4) A1 (AO3.4) [5]	Multiplying by IF and writing LHS as an exact derivative “+ c ” required Use of initial conditions to derive a value for c	Or CF Or subst correct PI into DE Or GS $v = Ae^{-t} - 2e^{-2t}$ Or using alternative boundary condition	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	c		As $t \rightarrow \infty$, $v \rightarrow 0$ So speed starts at 0 and ends at 0 (and is continuous and positive between) so must reach a maximum somewhere in $t > 0$	M1 (AO3.4) A1 (AO2.4) [2]		
	d		v is max when $\frac{dv}{dt} = 0$ so $t = \ln 2$ so $t = \ln 2$ So $v_{\max} = 0.5$ (given) (or $v_{\max} = 2e^{-\ln 2} - 2e^{-2\ln 2} = 1 - \frac{2}{4} = \frac{1}{2}$)	M1 (AO2.2a) A1 (AO3.4) [2]	Deducing time when v is maximum Or by finding expression for $\frac{dv}{dt}$ and solving $\frac{dv}{dt} = 0$	
	e		$v = \frac{dx}{dt} = 2e^{-t} - 2e^{-2t} \Rightarrow x = -2e^{-t} + e^{-2t} + d$ $t = 0, x = 0 \Rightarrow 0 = -2 + 1 + d \Rightarrow d = 1$ $0.9 = -2e^{-t} + e^{-2t} + 1$ $(e^{-t})^2 - 2e^{-t} + 0.1 = 0 \Rightarrow e^{-t} = \frac{10 \pm 3\sqrt{10}}{10}$ $\Rightarrow t = \ln\left(\frac{10}{10 - 3\sqrt{10}}\right) = 2.97$ (3 sf)	M1 (AO3.3) M1 (AO3.3) M1 (AO3.5a) A1 (AO2.3) [4]	Integrating to find expression for x Using initial conditions to find value of (new) constant Recognising that the model is only valid when x lies between 0 and 0.9 Rejecting $t = \ln\left(\frac{10}{10 + 3\sqrt{10}}\right) < 0$ (can be implicit) Or definite integral with correct lower limit.. ...and upper limit	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	16	

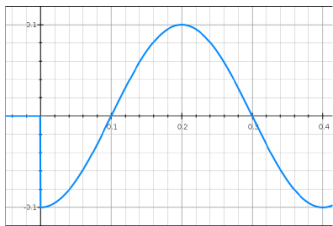


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Question		Answer/Indicative content	Marks	Part marks and guidance	
3	a		B1(AO1.2)B1(AO3.4)B1(AO3.1b)[4]	At least one cycle of $A \cos \omega t$ graph Amplitude 0.1 Period 0.4 Intersect with the t -axis at 0.1 and 0.3 (values must be indicated or implied unambiguously (eg by tick-marks and a single value))	graph must be instantaneous / horizontally at top / bottom, continuous and not vertical at any point. Ignore any graph outside $[0, 0.4]$ Non-inverted cos graph can still get 4/4
				<u>Examiner's Comments</u> In answering this question a cosine graph for $x(t)$ was often (correctly) selected, although the sine graph was seen. Most candidates did show a single cycle of their wave function with the correct period 0.4s, however very few candidates understood that the displacement at $t = 0$ was negative and draw the correct 'inverted' cosine graph. While the lack of 'inversion' was not penalised at this stage, it did lead to an incorrect value for the displacement in part (b). Some candidates gave the amplitude as 0.2m instead of 0.1m and many neglected to label the intersections with the t axis	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					at 0.1s and 0.3s.	
	b		So $x = \pm 0.1 \cos(5\pi t)$ or $x = 0.1 \sin(5\pi t \pm \frac{1}{2}\pi)$ when $t = 0.75$ or 0.35 $x = -\frac{\sqrt{2}}{20}$ (= -0.0707 to 3 sf)	M1(AO3.1 b) A1(AO3.4) [2]	or by argument from sketch Condone amplitude of 0.2 for M1 <u>Examiner's Comments</u> Many candidates were able to find $\omega = 5\pi$, usually gaining one mark for the absolute displacement. A few started from the general expression $a\cos(\omega t) + b\sin(\omega t)$ for $x(t)$, despite having drawn a cosine graph in part (a) and these usually did not arrive at the conclusion that $b = 0$.	
			Total	6		



Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	$F = ma \Rightarrow \frac{1}{2}t - v = 2 \frac{dv}{dt} \Rightarrow \frac{dv}{dt} + \frac{1}{2}v = \frac{1}{4}t$	B1(AO3.3)[1]	AG. Intermediate step as shown or both sides halved		
				<u>Examiner's Comments</u> In order to show how to derive the given differential equation, candidates were required to start by quoting $F = ma$ before using it. A small number of candidates' working showed a conceptual misunderstanding, from seemingly believing that there could be two accelerations involved.		
	b	$IF = e^{\int \frac{1}{2} dt} = e^{\frac{1}{2}t}$ $e^{\frac{1}{2}t} \frac{dv}{dt} + \frac{1}{2}e^{\frac{1}{2}t} v = \frac{d}{dt} \left(v e^{\frac{1}{2}t} \right) = \frac{1}{4} t e^{\frac{1}{2}t}$ $v e^{\frac{1}{2}t} = \frac{1}{2} t e^{\frac{1}{2}t} - \frac{1}{2} \int e^{\frac{1}{2}t} dt = \frac{1}{2} t e^{\frac{1}{2}t} - e^{\frac{1}{2}t} (+c)$ $0 = -1 + c \Rightarrow c = 1$ $v = \frac{1}{2}t - 1 + e^{-\frac{1}{2}t}$ Alternate method $AE: \lambda + \frac{1}{2} = 0, \lambda = -\frac{1}{2}$ $CF: (v =) A e^{-\frac{1}{2}t}$ Trial function: $v = at + b, \frac{dv}{dt} = a$	B1(AO1.1) M1(AO1.1) A1(AO1.1) M1(AO3.4) A1(AO1.1) B1 M1 A1 M1 A1 [5]	$\text{Or } e^{\frac{1}{2}t+c} \text{ or } Ae^{\frac{1}{2}t}$ Multiplying both sides by IF and writing new LHS as an exact derivative Use initial conditions to determine c Substitutes	Allow if using a solution of correct form with wrong (non-zero) coefficients Allow if using a solution of correct	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$a + \frac{1}{2}(at + b) = \frac{1}{4}t$ $a = \frac{1}{2}, b = -1$ GS: $(v =) Ae^{-\frac{1}{2}t} + \frac{1}{2}t - 1$ $v = 0, t$ gives $A = 1$ $v = e^{-\frac{1}{2}t} + \frac{1}{2}t - 1$		boundary condition into general solution to find A form with wrong (non-zero) coefficients <u>Examiner's Comments</u> Most candidates attempted to use an integrating factor (IF) and could usually find the correct one. Marks were lost due to · not multiplying both sides by the IF, · an inability to integrate $te^{t/2}$ by parts, · poor handling of the constant of integration, often leading to the erroneous form $v = f(x) + c$. These problems did not present themselves to candidates who solved the equation by finding the complementary function (CF) and a particular integral (PI). Some who took this equally valid approach did not however know the most efficient trial function to use in order to discover their PI.		
			$t = 2 \Rightarrow v = e^{-1}$ (so velocity is $e^{-1} \text{ ms}^{-1}$)	B1(AO3.4)[1]	or awrt 0.368 <u>Examiner's Comments</u> Virtually all the candidates who obtained the correct equation for v in part (a) were able to find the value of the velocity at $t = 2$, mostly preferring the exact value e^{-1} .		



Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		change the RHS of DE in part (a) from $\frac{1}{4}t$ to $\frac{1}{4}$ oe	B1(AO3.5c)[1]		<u>Examiner's Comments</u> Most candidates derived the new equation of motion successfully. A few candidates however read the unit symbol for Newton in $\frac{1}{2}N$ as a constant and inserted $\frac{N}{4}$ as the constant term in their differential equation. Had they followed the example given in part (a), they should have been able to correct their error.



Question		Answer/Indicative content	Marks	Part marks and guidance	
e		$\frac{d}{dt}\left(v e^{\frac{1}{2}t}\right) = \frac{1}{4} e^{\frac{1}{2}t} \Rightarrow v e^{\frac{1}{2}t} = \int \frac{1}{4} e^{\frac{1}{2}t} dt$ $v e^{\frac{1}{2}t} = \frac{1}{2} e^{\frac{1}{2}t} + c \quad \text{oe}$ $e^{-1} e^1 = \frac{1}{2} e^1 + c \Rightarrow c = 1 - \frac{1}{2} e$ $v = \frac{1}{2} + \left(1 - \frac{1}{2} e\right) e^{-\frac{1}{2}t}$	B1(AO2.2a) M1(AO3.3) A1(AO1.1)[3]	S $v e^{\frac{1}{2}t} = \frac{1}{2} e^{\frac{1}{2}t} + c$ oe C1 substitute their boundary condition from (c) into correct GS to find c <u>Examiner's Comments</u> Most candidates did not take advantage of their prior work but started to solve the refined differential equation from scratch. The most efficient course of action was to recognise the effect of an unchanged term in v. A significant number of candidates opted for the separation of variables method, which led to a more complicated procedure when finding v explicitly. Some also applied the incorrect boundary condition at t = 0 when attempting to find the new arbitrary constant.	$\frac{dv}{dt} + \frac{1}{2} v = \frac{1}{4}$ rearranged to $\int \frac{2dv}{1-2v} = \frac{1}{2} \int dt = \frac{1}{2} t$ giving $-\ln(1-2v) + c' = \frac{1}{2} t$ $c' = 1 + \ln(1-2e^{-1})$
Total			11		



Question		Answer/Indicative content	Marks	Part marks and guidance		
5	a	AE: $m^2 - 2m - 15 = 0 \Rightarrow m = 5 \text{ or } -3$ So GS is $Q = Ae^{-3t} + Be^{5t}$ Q tends to finite limit as $t \rightarrow \infty \Rightarrow B = 0$ $t = 0, Q = 100 \Rightarrow A = 100$ So $Q = 100e^{-3t}$ $t = 0.5 \Rightarrow Q = 100e^{-1.5} = 22.3$	M1 (AO1.1) A1 (AO1.1) B1 (AO2.2a) M1 (AO3.4) A1 (AO1.1) A1 (AO3.4) [6]	Or dQ/dt tends to zero as $t \rightarrow \infty$ Using initial condition to find A (or $A + B$) soi	www	
	b	(As $t \rightarrow \infty e^{-3t} \rightarrow 0$ so Q tends to) 0.	B1 (AO3.4) [1]	Only if from $Q = ke^{-at}$, $a > 0$	It must be clear that the limit is 0; "Q is approximately 0" would not be sufficient for B1.	
		Total	7			



Question			Answer/Indicative content	Marks	Part marks and guidance	
6	a		$F = ma$ $(F_1 \text{ is in the direction of motion and } F_2 \text{ is resisting motion})$ $F = \lambda t - \mu v$ $\Rightarrow \frac{1}{2} \frac{dv}{dt} = \lambda t - \mu v$ AG	M1 (AO3.3) A1 (AO2.1) [2]	Use of Newton II with constants of proportionality. $F = ma$ must be seen	
	b		$\frac{1}{2} \frac{dv}{dt} = \lambda t - \mu v$ $\Rightarrow \frac{dv}{dt} + 2\mu v = 2\lambda t$ I.F. $e^{2\mu t}$ $\Rightarrow e^{2\mu t} \frac{dv}{dt} + e^{2\mu t} 2\mu v = 2\lambda t e^{2\mu t}$ $\Rightarrow \frac{d}{dt}(e^{2\mu t} v) = 2\lambda t e^{2\mu t}$ $\Rightarrow e^{2\mu t} v = \int 2\lambda t e^{2\mu t} dt$ $\Rightarrow e^{2\mu t} v = 2\lambda \left(\frac{1}{2\mu} t e^{2\mu t} - \frac{1}{4\mu^2} e^{2\mu t} \right) + c$ Given that $v = 0 \Rightarrow c = \frac{2\lambda}{4\mu^2}$ $t = 0,$ $\Rightarrow v = \frac{\lambda}{\mu} t - \frac{\lambda}{2\mu^2} + \frac{\lambda}{2\mu^2} e^{-2\mu t}$ oe Alternative method $\frac{dv}{dt} + 2\mu v = 2\lambda t$ $AE : m + 2\mu = 0$ $\Rightarrow m = -2\mu \Rightarrow CF = Ae^{-2\mu t}$ $PI : v = at + b$	M1* (AO1.1a) A1 (AO1.1) M1 dep (AO3.1a) A1 (AO1.1) M1 dep (AO3.1a) A1 (AO1.1) A1 (AO3.4)	Attempt to find IF Getting DE in correct form Attempt integration by parts	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow \frac{dv}{dt} = a \Rightarrow a + 2\mu at + 2\mu b = 2\lambda t$ $\therefore a = \frac{\lambda}{\mu}, b = -\frac{a}{2\mu} = -\frac{\lambda}{2\mu^2}$ $\therefore GS: v = \frac{\lambda}{\mu}t - \frac{\lambda}{2\mu^2} + Ae^{-2\mu t}$	M1 (AOC) A1 (AOC) A1 (AOA) [7]		
	c		$v = \frac{\lambda}{\mu}t - \frac{\lambda}{2\mu^2} + \frac{\lambda}{2\mu^2}e^{-2\mu t}$ $\lambda = 2, \mu = 1 \Rightarrow v = 2t - 1 + e^{-2t}$ When t is large, e^{-2t} is very small so $v \approx 2t - 1$	M1 (AO3.4) A1 (AO3.3) [2]	Consider the behaviour of the exponential function in <i>their</i> equation from (b) so $v \approx 2t$	
	d		$\frac{1}{2} \frac{dv}{dt} = 2 - v$	B1 (AO3.5c) [1]		

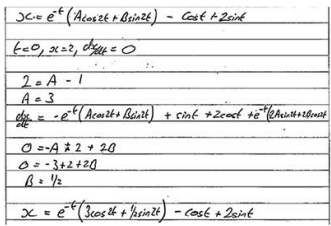


Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		As $v \rightarrow 0$ approaches 2, i.e. v approaches a constant value.	B1 (AO3.4) [1]	<u>Examiner's Comments</u> The differential equation in part (a) was a statement of Newton's second law. In order to complete the "show that" this had to be stated. Not many candidates did this and some responses lacked the mathematical argument required to justify the given answer. In part (b) the integrating factor was usually found and an integration by parts performed satisfactorily. There were many steps along the way, however, including the finding of the arbitrary constant of integration. This meant that candidates had to interpret the initial conditions from the stem of the question. For those who found the correct particular solution the approximation in part (c) was straightforward. Those that had a particular solution that did not include the exponential term might have accepted a clue that something was wrong here. Part (d) for the refined model was usually answered well but part (e) had very mixed responses.	
			Total	13		



Question		Answer/Indicative content	Marks	Part marks and guidance		
7	a	$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 10\sin t$ A.E $n^2 + 2n + 5 = 0 \Rightarrow n = -1 \pm 2i$ $\Rightarrow x = e^{-t}(A \cos 2t + B \sin 2t)$ P.I. $x = a \sin t + b \cos t$ $\Rightarrow \frac{dx}{dt} = a \cos t - b \sin t$ $\Rightarrow \frac{d^2x}{dt^2} = -a \sin t - b \cos t$ $\Rightarrow -a \sin t - b \cos t + 2(a \cos t - b \sin t) + 5(a \sin t + b \cos t) = 10 \sin t$ $\Rightarrow -b + 2a + 5b = 0, -a - 2b + 5a = 10$ i.e. $a = -2b, 2a - b = 5 \Rightarrow b = -1, a = 2$ $\Rightarrow$ G.S. $x = e^{-t}(A \cos 2t + B \sin 2t) + 2 \sin t - \cos t$ When $t = 0$ we have $x = 2$ $\text{and } \frac{dx}{dt} = 0$ $\Rightarrow A - 1 = 2 \Rightarrow A = 3$ $\frac{dx}{dt} = -e^{-t}(A \cos 2t + B \sin 2t) + e^{-t}(-2A \sin 2t + 2B \cos 2t) + 2 \cos t + \sin t$ $t = 0 \Rightarrow -A + 2B + 2 = 0 \Rightarrow$ $B = \frac{1}{2}$ $\Rightarrow x = e^{-t} \left(3 \cos 2t + \frac{1}{2} \sin 2t \right) + 2 \sin t - \cos t$	M1(AO1. 1a) A1(AO1. 1) M1(AO3. 1a) M1(AO1. 1) A1(AO1. 1) A1(AO1. 1) B1(AO3. 3) B1(AO3. 3) M1(AO3. 4) A1(AO3. 3) A1(AO3. 4) [11]	For obtaining the AE Solving AE and interpreting Correct form of PI and differentiate twice Dep on previous M Substitute their PI with two constants into DE Values of a and b Correct GS For substituting into 2 eqns For A Dep on previous M For diffn their GS. Their value for A could be substituted For B	Or equivalent forms (i.e. exponential form)	<u>Examiner's Comments</u> This was a long question



Question			Answer/Indicative content	Marks	Part marks and guidance	
					with a number of parts to it. Candidates should be able to write their working clearly and logically (both were sadly lacking in a number of scripts). The complementary function was usually found, although a number of candidates wrote $x = e^t (A \cos 2t + B \sin 2t) \text{ or } x = e^{-1} (A \cos 2t + B \sin 2t).$ A small number of candidates then used the initial and boundary conditions to find the parameters before adding the particular integral. There was no complication with the particular integral (in that the function in the differential equation did not form part of the complementary function), but even so a few candidates took their particular integral to be $t(p \cos t + q \sin t)$. Exemplar 3  This candidate found the general solution and then differentiated appropriately to use the boundary conditions to find the particular solution. At this stage of the question it was disappointing to see so many candidates not	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					differentiate using the product rule, which at this level should be a basic skill.	
	b		$x = e^{-t} (A \cos 2t + B \sin 2t) + 2 \sin t - \cos t$ As $t \rightarrow \infty$, $e^{-t} \rightarrow 0$ $\Rightarrow x \approx 2 \sin t - \cos t$	M1(AO3.4) A1(AO3.4) [2]	Consider behaviour of e^{-t} Accept = ft <i>their</i> equation from (a)	Accept this final line for 2 marks
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a	$\frac{dx}{dt} = y \Rightarrow \frac{d^2x}{dt^2} = \frac{dy}{dt}$ $= 2y - 5x$ $\Rightarrow \frac{d^2x}{dt^2} = 2 \frac{dx}{dt} - 5x$	M1 (AO 3.4) A1 (AO 2.1) [2]	Differentiate 1st DE and substitute back in AG	
		b	(i) $\frac{d^2x}{dt^2} - 2 \frac{dx}{dt} + 5x = 0$ A.E. is $n^2 - 2n + 5 = 0$ $\Rightarrow n = 1 \pm 2i$ $\Rightarrow x = e^t (A \sin 2t + B \cos 2t)$ (ii) $x = e^t (A \sin 2t + B \cos 2t)$ and $\frac{dx}{dt} = y$ $\Rightarrow \frac{dx}{dt} = e^t (A \sin 2t + B \cos 2t) + e^t (2A \cos 2t - 2B \sin 2t)$ $-y = e^t ((A - 2B) \sin 2t + (2A + B) \cos 2t)$	M1 (AO 1.1) A1 (AO 1.1) A1 (AO 2.2a) [3] M1 (AO 1.1) A1 (AO 2.2a) [2]	For auxiliary equation BC	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$x = e^t(A \sin 2t + B \cos 2t)$ $x = 1, t = 0$ $\Rightarrow 1 = B$ $\Rightarrow y = e^t((A - 2)\sin 2t + (2A + 1)\cos 2t)$ $y = 3, t = 0$ $\Rightarrow 3 = 2A + 1 \Rightarrow A = 1$ $\Rightarrow x = e^t(\sin 2t + \cos 2t), y = e^t(3 \cos 2t - \sin 2t)$	M1 (AO 3.3) A1 (AO 1.1) M1 (AO 3.3) A1 (AO 1.1) A1 (AO 3.3) [5]	Translate context into condition to find B Translate context into condition to find A Both stated with A and B substituted for	
		d	$x = 0 \Rightarrow x = e^t(\sin 2t + \cos 2t) = 0$ $t = 1.178...$ So 1.12 year $y = 0 \Rightarrow y = e^t(3 \cos 2t - \sin 2t) = 0$ $t = 0.624...$ So 0.625 year 0.625 < 1.12 so the prey die out first.	M1 (AO 3.1b) A1 (AO 3.4) A1 (AO 3.4) E1 (AO 3.4) [4]	Set either their $x = 0$ or their $y = 0$ and solve BC BC	
			Total	16		



Question			Answer/Indicative content	Marks	Part marks and guidance		
9		a	(i) $\frac{d^2\theta}{dt^2} = -\left(\frac{5}{2}\right)^2 \theta$ $\theta = A \cos \omega t + B \sin \omega t$ or $R \cos(\omega t + \phi)$ $\theta = A \cos \frac{5}{2}t + B \sin \frac{5}{2}t$ or $R \cos\left(\frac{5}{2}t + \phi\right)$ (ii) The model predicts infinite oscillations of the same amplitude; in practice the amplitude must decrease over time.	M1 (AO 1.1) A1 (AO 1.1) [2] E1 (AO 3.5b) [1]			If M0 then SC1 for $\theta = A \sin \frac{5}{2}t$ or $\theta = A \sin \frac{5}{2}t$
		b	(i) AE: $4m^2 + 16m + 25 = 0$ $-2 \pm \frac{3}{2}i$ $\theta = e^{-2t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right)$ (ii) $t = 0, \theta = 0.9 \Rightarrow A = 0.9$ $\frac{d\theta}{dt} = -2e^{-2t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right) + e^{-2t} \left(-\frac{3}{2}A \sin \frac{3}{2}t + \frac{3}{2}B \cos \frac{3}{2}t \right)$ $t = 0, \frac{d\theta}{dt} = 0 \Rightarrow -2A + \frac{3}{2}B = 0$	M1 (AO 1.1a) A1 (AO 1.1) A1ft (AO 1.1) [3] B1 (AO 3.4) M1 (AO 1.1a) M1 (AO	Writing down the AE correctly or using $\theta = Ae^{mt}$ and substituting into (*) to derive a three term quadratic AE. Their ept ($A \cos qt + B \sin qt$) for solution of AE = $p \pm qi$ Attempt to differentiate		



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$B = 1.2$ $\theta = e^{-2t} \left(0.9 \cos \frac{3}{2}t + 1.2 \sin \frac{3}{2}t \right)$ (iii In the modified model $\theta \rightarrow 0$ as $t \rightarrow \infty$ or This is the behaviour we would expect to observe with a real swing door and so the model is an improvement.	3.4) A1 (AO 1.1) [4] B1 (AO 3.5a) B1 (AO 3.5c) [2]	using the product and chain rules (A may be replaced by a number). Substituting $t = 0$ into $\frac{d\theta}{dt}$ to derive an equation in (A and) B ie the amplitude decays etc		
		c	Need $4m^2 + \lambda m + 25 = 0$ to have repeated solutions so $\lambda^2 - 4 \times 4 \times 25 = 0$ $\lambda > 0 \Rightarrow \lambda = 20$	M1 (AO 3.5c) A1 (AO 3.3) [2]	Using "b ² - 4ac" = 0 directly Not -20 or ± 20	or $\left(2m + \frac{\lambda}{4} \right)^2 + 25 - \frac{\lambda^2}{16} = 0$ and equating part outside brackets to 0	
			Total	14			



Question			Answer/Indicative content	Marks	Part marks and guidance		
10		a	$\frac{dx}{dt} + 0.1x = \cos t e^{-0.1t} \Rightarrow \text{I.F} = e^{0.1t}$ $\Rightarrow e^{0.1t} \times \frac{dx}{dt} + 0.1e^{0.1t} \times x = \cos t$ $\Rightarrow \frac{d}{dx}(e^{0.1t} x) = \cos t$ $\Rightarrow e^{0.1t} x = \sin t + c \Rightarrow x = (\sin t + c)e^{-0.1t}$	B1 (AO3.3) M1 (AO3.4) A1 (AO3.4) [3]	Multiplying throughout by IF and writing RHS as an exact derivative of a product		
		b	$x = (\sin t + c)e^{-0.1t}; x = 10, t \Rightarrow c = 10$ $x = (\sin t + 10)e^{-0.1t}$	M1 (AO3.4) A1 (AO3.4) [2]	Substituting $x = 10$ and $t = 0$ to find a value for c		
		c	(i 5.33477) (ii 5.33486)	B1 (AO3.4) [1] B1 (AO3.4) [1]	to 6sf or better to 6sf or better		
		d	E.g. the amount at 6.25 hours is higher than at 6 hours, so the amount is not always decreasing, and so model is not appropriate (for predicting decay)	B1 (AO 3.5a) [1]	Conclusion required		
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	(i) $\frac{d^2x}{dt^2} = 3\frac{dx}{dt} - 2\frac{dy}{dt} = 3\frac{dx}{dt} - 2(y + 5x)$ $\frac{d^2x}{dt^2} = 3\frac{dx}{dt} - \left(3x - \frac{dx}{dt}\right) - 10x = 4\frac{dx}{dt} - 13x$ (ii) $m^2 - 4m + 13 = 0$ $m = 2 \pm 3i$ $x = e^{2t} (A \sin 3t + B \cos 3t)$	M1 (AO 2.1) A1 (AO 2.2a) [2] M1 (AO 3.3) A1 (AO 1.1) A1ft (AO 3.3) [3]	Differentiating $\frac{dx}{dt} = 3x - 2y$ and substituting in for $\frac{dy}{dt}$ from $\frac{dy}{dt} = y + 5x$ Substituting for a second time to leave an equation in x, expanding and collecting. No need to see $a = 4$, $b = -$ explicitly Finding the auxiliary equation and attempting to solve Can be implied by correct form	Attempt to solve can be implied either by correct solution or by working (eg $(m - 2)^2 + 9 = 0$) FT on complex solutions to the AE only.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$\frac{d^2 y}{dt^2} = \frac{dy}{dt} + 5 \frac{dx}{dt} = \frac{dy}{dt} + 5(3x - 2y)$ $\frac{d^2 y}{dt^2} = \frac{dy}{dt} + 3 \left(\frac{dy}{dt} - y \right) - 10y = 4 \frac{dy}{dt} - 13y$ Same form as for x, so $y = e^{2t} (C \sin 3t + D \cos 3t)$	M1 (AO 2.1) M1 (AO 2.2a) B1ft (AO 2.2a) [3]	Differentiating $\frac{dy}{dt} = y + 5x$ and $\frac{dx}{dt} = 3x - 2y$ substituting in for $\frac{dx}{dt}$ from $\frac{dx}{dt} = 3x - 2y$ Substituting for a second time to leave an equation in y , expanding and collecting. Arbitrary constants must not be "A" or "B" from (a)(ii)	No need to see $a = 4$, $b = -13$ or explicitly FT on their answer to (a)(ii) with two arbitrary constants	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$t = 0 \Rightarrow B = 4 \text{ \& } D = 5$ $\frac{dy}{dt} = 2e^{2t} (C \sin 3t + D \cos 3t)$ $+ e^{2t} (3C \cos 3t - 3D \sin 3t)$ $= y + 5x = e^{2t} (C \sin 3t + D \cos 3t)$ $+ 5e^{2t} (A \sin 3t + B \cos 3t)$ Comparing coefficients gives $2C - 3D = C + 5A$ $2D + 3C = D + 5B$ So $A = -2, C = 5$ Particular solutions are $x = 2e^{2t} (-\sin 3t + 2 \cos 3t)$ and $y = 5e^{2t} (\sin 3t + \cos 3t)$	B1 (AO 3.3) M1 (AO 3.4) M1 (AO 3.3) A1 (AO 1.1) A1 (AO 1.1) [5]	Attempt to differentiate expression for x or y using the product rule (could have B or D substituted out) Using the correct equation to set up equations in C and D (could have B or D substituted out)	$\frac{dx}{dt} = 2e^{2t} (A \sin 3t + B \cos 3t) + e^{2t} (3A \cos 3t - 3B \sin 3t)$ $= 3x - 2y = 3e^{2t} (A \sin 3t + B \cos 3t) - 2e^{2t} (C \sin 3t + D \cos 3t)$ $\Rightarrow 2A - 3B = 3A - 2C$ $\text{ \& } 2B + 3A = 3B - 2D$	
		d	$x = 0 \Rightarrow \tan 3T = 2$ $\Rightarrow T = 0.369$	B1 (AO 3.4) [1]	Setting $x = 0$ to and rearrange to an equation in $\tan 3T$	$0.369049...$	
		e	E.g. The model has continuous quantities, but the number of rabbits can only take integer values.	E1 (AO 3.5a) [1]			
			Total	15			



Question			Answer/Indicative content	Marks	Part marks and guidance	
12			$\frac{dQ}{dt} = -k(1+Q^2) \Rightarrow \int \frac{dQ}{1+Q^2} = -k \int dt$ $\Rightarrow \tan^{-1} Q = c - kt$ $t = 0, Q = 100 \Rightarrow c = \tan^{-1} 100$ $t = 100, Q = 50$ $\Rightarrow \tan^{-1} 50 = \tan^{-1} 100 - 100k$ $\Rightarrow k = \frac{1}{100}(\tan^{-1} 100 - \tan^{-1} 50) \approx 0.0001$ When $t = 400, Q = 20$ 20 g	B1 (AO3.3) M1 (AO3.1b) A1 (AO1.1) A1 (AO3.4) A1 (AO3.4) M1 (AO1.1a) A1 (AO3.4) A1 (AO3.2b) [8]	Must have unit	
			Total	8		
13		a	$\theta = A \cos 2t + B \sin 2t$	B1 (AO1.2) [1]		
		b	$\theta = A \cos 2t + B \sin 2t$ $\Rightarrow \frac{d\theta}{dt} = -2A \sin 2t + 2B \cos 2t$ $\frac{d\theta}{dt} = 0$ when $t = 0 \Rightarrow B = 0$ $\theta = \theta_0$ when $t = 0 \Rightarrow A = \theta_0$ $\Rightarrow \theta = \theta_0 \cos 2t$	M1 (AO3.3) A1 (AO3.4) A1 (AO1.1) A1 (AO3.4) [4]		
		c	Angle has to be small otherwise the model is invalid.	B1 (AO3.5b) [1]		
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					others showed this when they needed to replace $\sin \theta$ by θ in their proof. Some candidates only realised the need to show this when they needed to find the amplitude in part (ii) and, in this case, credit was awarded in part (i) for the work shown in part (ii).	
		ii	$0.087266 = A \sin 3.5t$ $t = 0.238 \text{ secs}$ $t' = 2 \left(\frac{1.7952}{4} - 0.2378 \right) = 0.422(\text{s})$ $\dot{\theta} = 0.118 \times 3.5 \cos 3.5 \times 0.238$ Linear speed = $0.222(\text{ms}^{-1})$	M1 A1 A1 M1 A1 [5]	OR $5 = A \sin 3.5t$; $A =$ amplitude Or 0.65972 Or $0.65972 - 0.2378$ OR $\dot{\theta} = \sqrt{(3.5^2 (0.118^2 - 0.0873^2))} \times 0.278$ May use $\cos^*$ 0.2378 allow $\sin$ if consistent with $*$; allow 5° and 6.76° Or $\frac{1}{2} m \frac{9}{90} = \frac{1}{2} m v^2 + mg 0.8(1 - \cos 5^\circ)$	Examiner's Comments Candidates with the correct amplitude were usually able to find the times taken for the string to make an angle of 5° with the vertical, though not all found the time that elapsed between the two occasions. Candidates with the wrong amplitude were allowed partial credit for a correct method to find a time, and for a correct method to find the linear speed of P .



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Max height is still 0.00556 so i Max angle = $\cos^{-1} \frac{(0.05-0.00556)}{0.05}$ [27.3] Not SHM since angle is not small	B1 B1 B1 [3]	accept 'still the same' or attempt to work out height 0.476 rads Examiner's Comments A small number of candidates answered this very well, by finding the angle made with the vertical by the string when <i>P</i> at its highest point to the right of <i>Q</i> and explaining that this angle is too big for the approximation of $\sin \theta$ by θ to be valid. Many candidates gained partial credit for an answer stating that the angle was too big without showing any detailed working. Other candidates gave wrong answers referring to the string becoming slack, <i>P</i> moving in a circle or other forces being involved to suggest that the motion is not simple harmonic.	
		Total		15		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
15			AE: $\lambda^2 + 6\lambda + 9 = 0$ $\lambda = -3$ (repeated) CF: $(A + Bx)e^{-3x}$ PI: $y = a \cos x + b \sin x$ $y' = a \sin x + b \cos x$ $y'' = -a \cos x - b \sin x$ In DE: $-a \cos x - b \sin x + 6(-a \sin x + b \cos x) + 9(a \cos x + b \sin x) = 25 \sin x$ $-a + 6b + 9a = 0$ $-b - 6a + 9b = 25$ $a = -1.5, b = 2$ GS: $y = 2 \sin x - 1.5 \cos x + (A + Bx)e^{-3x}$	M1 A1 A1ft B1 M1 M1 A1 A1 [8]	CF for their roots (with two constants) Differentiate twice and substitute Compare coefficients PI correct		
Total				8	Examiner's Comments Many candidates were able to reach the correct solution with ease. The commonest errors included: ♦ $\lambda = 3$ leading to $y_c = (A + Bx)e^{3x}$ ♦ $y_c = Ae^{-3x}$ or $(A + B)e^{-3x}$ ♦ miscalculation of a, b in the particular integral ♦ omission of "$y = \dots$" in the final answer.		



Question			Answer/Indicative content	Marks	Part marks and guidance				
16			$(I \Rightarrow) \exp(\int \cot x \, dx)$	M1	Multiply and integrate				
			$= e^{\ln \sin x}$	M1					
			$= \sin x$	A1					
			$\frac{d}{dx}(y \sin x) = 9$	M1*					
			$y \sin x = 9x + A$	A1					
			$x = \frac{1}{6}\pi, y = \pi \Rightarrow \frac{1}{2}\pi = \frac{3}{2}\pi + A \Rightarrow A = -\pi$	M1	Correct substitution of given point and constant evaluated				
			$y = (9x - \pi) \operatorname{Cosec} x$	*M1dep					
				A1				Rearrange to isolate “y” or	Must have “y =”
				[8]					
			Examiner’s Comments Most candidates had been well prepared for solving differential equations. They recognised the need for an integrating factor and could apply this method. Most, also, ensured that their constant of integration was included in the process of rearrangement to reach the form “y = ...”.The most common error was miscalculation of the constant due to poor algebraic manipulation.						



Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	$\frac{d}{dt} \left(0.5 \frac{dm}{dt} + m \right) = k(80 - m)$ $0.5 \frac{d^2m}{dt^2} + \frac{dm}{dt} = k(80 - m)$ $\Rightarrow \frac{d^2m}{dt^2} + 2 \frac{dm}{dt} = 2k(80 - m)$ $\Rightarrow \frac{d^2m}{dt^2} + 2 \frac{dm}{dt} + 2km = 160k$	B1(AO 3.3) E1(AO 2.1) [2]	Translate the given context into a correct mathematical model AG		
		b	$\frac{d^2m}{dt^2} + 2 \frac{dm}{dt} + 2km = 160k$ $\Rightarrow \text{A.E. } n^2 + 2n + 2k = 0$ $\Rightarrow n = \frac{-2 \pm \sqrt{4 - 8k}}{2} = -1 \pm \sqrt{1 - 2k}$ CF in the form $A \cos + 2t + B \sin 2t \Rightarrow$ $1 - 2k < 0$ so $\sqrt{2k - 1} = 2 \Rightarrow k = \frac{5}{2}$ $\lambda = -1$	M1(AO 1.1) A1(AO 1.1) E1(AO 2.2a) A1ft(AO1.1) [4]	For auxiliary equation Need not be simplified AG Follow through their auxiliary equation	Or complete the square	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\frac{d^2m}{dt^2} + 2\frac{dm}{dt} + 5m = 400$ P.I. Try $m = \mu \Rightarrow \frac{dm}{dt} = \frac{d^2m}{dt^2} = 0$ Substituting $\Rightarrow 5\mu = 400 \Rightarrow \mu = 80$ $\Rightarrow$G.S. $m = e^{-t}(A\cos 2t + B\sin 2t) + 80$ $m = 0$ when $t = 0$ $\Rightarrow A = -80$ $\frac{dm}{dt} = 160$ when $t = 0$ $\frac{dm}{dt} = -e^{-t}(A\cos 2t + B\sin 2t) + 2e^{-t}(-A\sin 2t + B\cos 2t)$ $= 160$ when $t = 0$ $\Rightarrow -A + 2B = 160 \Rightarrow B = 40$ $\Rightarrow m = e^{-t}(40\sin 2t - 80\cos 2t) + 80$	M1(AO 3.1a) A1(AO 1.1) M1(AO 3.3) A1(AO 1.1) M1(AO 3.3) M1(AO 1.1) A1(AO 1.1) [7]	Correct Particular Integral and substituting General Solution including correct μ Translate the given context into an initial condition to find A Translate the given context into an initial condition to find B Differentiation and substitution Final answer stated	Award first two marks if GS stated correctly	
		d	As $t \rightarrow \infty$, $m \rightarrow 80$	B1(AO 3.4) [1]			



Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	E.g. There is a limiting process here. According to the model the mass of the crate is only displayed after infinite time	B1(AO 3.5b) [1]		
		f	(i) $m = e^{-t}(40\sin 2t - 80\cos 2t) + 80$ $\Rightarrow \frac{dm}{dt} = e^{-t}(160\cos 2t + 120\sin 2t) = 0$... when $t = 1.107 \Rightarrow m = 106$	M1(AO 3.1b) A1(AO 3.4) [2]	Differentiation, FT their expression from part (iii) Set 0 and solve BC, may be implied by 106	OR By graphical calculator M1 draw $m = f(t)$ A1 find value of m at the first maximum
		f	(ii) This is the maximum mass displayed (to 3 sf)	E1(AO 3.2a) [1]	Must state that this is the maximum value	
		g	e.g. $\frac{d^2m}{dt^2} + 2\frac{dm}{dt} + 5m = 500$	E1(AO 3.5c) [1]	Any differential equation of the form $\frac{d^2m}{dt^2} + 2\frac{dm}{dt} + am = b$ with $\frac{b}{a} = 100$	Accept any sensible suggestion
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Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	$A = 10$ $x = Ae^{kt} \Rightarrow \frac{dx}{dt} = Ake^{kt}$ giving $Ak = -4$ $10k = -4$ so $k = -0.4$	B1(AO3.3) M1(AO3.3) E1(AO1.1) [3]	Differentiate and use $\frac{dx}{dt} = \pm 4$ when $t = 0$ AG		
		b	(i) $\frac{dx}{dt} + \frac{dy}{dt} + \frac{dz}{dt} = -4e^{-0.4t}$ $+ (0.3x - 0.2y) + (0.2y + 0.1x)$ $= -0.4x + 0.3x + 0.1x = 0$	M1(AO3.4) E1(AO1.1) [2]	AG		
		b	(ii) $\frac{d}{dt}(x + y + z) = 0 \Rightarrow x + y + z = \text{constant}$ Throughout the industrial process the total amount of the three substances is constant	B1(AO2.2a) E1(AO3.4) [2]	i.e. no loss or gain of amounts Must be in context		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\frac{dy}{dt} + 0.2y = 3e^{-0.4t}$ $\times e^{0.2t}$ $ye^{0.2t} = 3 \int e^{-0.2t} dt$ $y = -15e^{-0.4t} + ce^{-0.2t}$ $c = 15$	M1(AO3.4) M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO1.1) [5]	Substitute x and rearrange Alternative CF and PI Allow arithmetic slips in coefficients of exponential terms and / or sign errors		
		d	$-3e^{-0.2t} + 6e^{-0.4t} = 0$ $t = 3.466$ $y_{\max} = 3.75 \text{ kg}$	M1(AO3.4) A1(AO1.1) A1(AO3.2a) [3]	FT their $\frac{dy}{dt}$ so BC All three marks may be implied by 3.75 kg BC	OR M1 $0.3x = 0.2y$ so $0.3 \times 10e^{-0.4t} = 0.2 \times (-15e^{-0.4t} + 15e^{-0.2t})$ A1 $t = 3.466$ BC A1 $y_{\max} = 3.75 \text{ kg}$ BC	
		e	9 kg of substance Z implies $x + y = 1$ $t = 13.4(25) \text{ hours}$ to 3 s.f. or better	M1(AO3.4) A1(AO1.1) [2]	Or by $z = 10 + 5e^{-0.4t} - 15e^{-0.2t} = 9$ BC		
			Total	17			



Question	Answer/Indicative content	Marks	Part marks and guidance	
19	AE: $\lambda^2 + 2\lambda + 10 = 0$ $\lambda = -1 \pm 3i$ CF: $e^{-x}(A \cos 3x + B \sin 3x)$ PI: $y = a \cos x + b \sin x$ $y' = -a \sin x + b \cos x$ $y'' = -a \cos x - b \sin x$ In DE: $-a \cos x - b \sin x + 2(-a \sin x + b \cos x) + 10(a \cos x + b \sin x) = 85 \cos x$ $-a + 2b + 10a = 85$ $-b - 2a + 10b = 0$ $a = 9, b = 2$	B1 B1 B1ft B1 M1* M1* A1	condone $Ae^{(-1+3i)x} + Be^{(-1-3i)x}$ Differentiate twice and substitute Compare coefficients PI correct	ft on complex λ only trial function $y = a \cos x$ scores max of B0 M1 M0 at this stage



Question			Answer/Indicative content	Marks	Part marks and guidance	
			GS: $y = 9 \cos x + 2 \sin x + e^{-x}$ ($A \cos 3x + B \sin 3x$)	*A1ft	Their CF (of standard form) + their PI	dep on gaining both M1 marks Examiner's Comments Most candidates had been well prepared for solving differential equations. The most common errors occurred in calculations made whilst solving the simultaneous equations. The ability to use a calculator which solves simple equations would obviously have been helpful to those prone to this type of error. Other occasional errors seen in scripts included using a complementary function involving a term in $\sin 3x$ or e^{-1} ($A \cos x + B \sin x$) and using the trial function $y = a \cos x$. There were still some occasions where candidates omitted "$y =$" from their general solution. Some candidates used θ instead of x and hence lost marks. A few left the CF in complex exponential form: the Examiner was keen that candidates evidence their understanding that a real solution exists.
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
20			AE: $\lambda^2 + 4\lambda + 13 = 0$	M1		
			$\lambda = -2 \pm 3i$	A1		
			CF: $e^{-2x}(A\cos 3x + B\sin 3x)$	A1ft	condone $Ae^{(-2+3i)x} + Be^{(-2-3i)x}$	ft on complex λ only
			PI: $y = a\cos x + b\sin x$	B1		
			$y' = -a\sin x + b\cos x$, $y'' = -a\cos x - b\sin x$ in DE: $-a\cos x - b\sin x + 4(-a\sin x + b\cos x) + 13(a\cos x + b\sin x) = \sin x$	M1	Differentiate twice and substitute	If wrong trial function can only gain a maximum of the next M1 and must use correct method to differentiate it
			$12a + 4b = 0$	M1	Compare	
			$12b - 4a = 1$			
			$a = -\frac{1}{40}$, $b = \frac{3}{40}$	A1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
			GS: $y = \frac{1}{40}(3\sin x - \cos x) + e^{-2x}(A\cos 3x + B\sin 3x)$	A1ft	<u>Examiner's Comments</u> Most candidates had been well-prepared for solving differential equations. The most common errors occurred in calculations made whilst solving the simultaneous equations. The ability to use a calculator, which solves simple equations, would obviously have been helpful to those prone to this type of error. Another, not infrequent, error occurred through poor transcription from complementary function to general solution. Other occasional errors seen included using a complementary function of the form $\cos 3x + i\sin 3x$ or $e^{-2}(A\cos x + B\sin x)$ and using trial functions $y = ax\cos x + bx\sin x$ or $y = a\sin x$. It was rarer, this year, for candidates to omit "y =" from their general solution. However, expressions beginning G.S. = y = ... were sometimes seen; although not penalised on this occasion, candidates should be encouraged to be precise in their use of the equals symbol.	ft must be of form $y = \text{"their CF + PI"}$ and of form "$a\cos x + b\sin x$ with a or b nonzero plus standard CF form" with 2 constants and not in complex exponential form
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
21			$\frac{dy}{dx} + \frac{3}{x}y = x + 1$	B1	Divide both sides by x	A0 means no further marks can be gained
			$I = \exp\left(\int \frac{3}{x} dx\right) = e^{3\ln x}$	M1		
			$= x^3$	A1		
			$x^3 \frac{dy}{dx} + 3x^2y = x^4 + x^3$			
			$\frac{d}{dx}(x^3y) = \dots$	M1	Multiply and recognise derivative	
			$\dots = x^4 + x^3$	M1	Integrate both sides (their two term polynomial)	condone absent A at this stage

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Question			Answer/Indicative content	Marks	Part marks and guidance	
			$x^3y = \frac{1}{5}x^5 + \frac{1}{4}x^4 + A$ $x=1, y=1 \Rightarrow A = \frac{11}{20}$ $y = \frac{1}{5}x^2 + \frac{1}{4}x + \frac{11}{20}x^{-3}$	A1 M1 A1	Use condition <u>Examiner's Comments</u> Students were, in general, very well prepared on the topic of differential equations, with plenty of them following the standard approach outlined in the markscheme. A few also successfully used CF+PI method, although some who took this approach erroneously formed an auxiliary equation for a homogeneous equation without constant coefficients. The most common errors were failing initially to divide the RHS by x and writing $\frac{A}{x^3} = k$ (a new constant). A few candidates falsely gave the LHS, after multiplication by IF, as $\frac{d}{dx}(3x^2y)$.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
22			AE: $\lambda^2 + 5\lambda + 6 = 0$ $\lambda = -2, -3$ CF: $Ae^{-2x} + Be^{-3x}$ PI: $y = ae^{-x}$ $ae^{-x} - 5ae^{-x} + 6ae^{-x} = e^{-x}$ $2a = 1$ $a = \frac{1}{2}$ GS: $(y =) \frac{1}{2}e^{-x} + Ae^{-2x} + Be^{-3x}$ $x = 0, y = 0 \Rightarrow \frac{1}{2} + A + B = 0$ $y' = -\frac{1}{2}e^{-x} - 2Ae^{-2x} - 3Be^{-3x}$ $x = 0, y' = 0 \Rightarrow -\frac{1}{2} - 2A - 3B = 0$ $A = -1, B = \frac{1}{2}$	B1 B1ft B1ft M1 A1 A1ft M1 M1* M1dep*	Differentiate and substitute Use condition on GS Differentiate their GS of form $y = k e^{-x} + Ae^{mx} + Be^{nx}$ where k, m, n are non-zero constants and m, n not 1 Use condition and attempt to find A, B	ft must be of form “ $k e^{-x}$ plus a standard CF form” with 2 arbitrary constants Must have 2 arbitrary constants

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Question			Answer/Indicative content	Marks	Part marks and guidance	
			$y = \frac{1}{2}e^{-x} - e^{-2x} + \frac{1}{2}e^{-3x}$	A1	www Examiner's Comments This question proved straightforward to many candidates. Marks were mainly lost due to calculation errors and again it is worth stressing the need for extra caution with calculations in a multistage problem involving so many marks. The place where most calculation errors crept in was in the finding of the arbitrary constants through solving two simple simultaneous equations. However, some candidates went wrong right at the start with their solutions to the quadratic, auxiliary equation. The most common method error occurred when candidates selected an unsuitable trial function for the particular integral.	Must have 'y ='
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
23			AE: $\lambda^2 + 2\lambda + 5 = 0$	M1		
			$\lambda = -1 \pm 2i$	A1		
			CF: $e^{-x}(A\cos 2x + B\sin 2x)$	A1ft		Or $Ae^{-x}\cos(2x + \alpha)$ Must be in real form
			PI: $y = ae^{-x}$	B1		If PI $y = axe^{-x}$, then max of M1, A1, A1, B0, M1, A0, A0 (since cannot be consistent) M1, M1, A1. Must have a constant in "their PI"
			$ae^{-x} - 2ae^{-x} + 5ae^{-x} = e^{-x}$ $4a = 1$ $a = \frac{1}{4}$	M1 A1	Differentiate & substitute	
			GS: $y = e^{-x}\left(\frac{1}{4} + A\cos 2x + B\sin 2x\right)$	A1ft		Must have "y ="
			$\frac{dy}{dx} = -e^{-x}\left(\frac{1}{4} + A\cos 2x + B\sin 2x\right)$ $+e^{-x}(-2A\sin 2x + 2B\cos 2x)$	M1*	Differentiate their GS of form $y = e^{-x}(P + A\cos nx + B\sin nx)$ where P is constant or linear term, n not 0 or 1	Allow one error
			$x=0, \frac{dy}{dx}=0 \Rightarrow -\left(\frac{1}{4} + A\right) + 2B = 0$			
			$x=0, y=0 \Rightarrow \frac{1}{4} + A = 0$	*M1	Use conditions	Must have 'y =' and be in real form
			$A = -\frac{1}{4}, B = 0$	A1ft	From their GS	
			$y = \frac{1}{4}e^{-x}(1 - \cos 2x)$	A1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					Examiner's Comments Given the fact that this question required standard steps, it was disappointing that the majority were unable to gain full marks. Almost all candidates knew how to approach the problem, but the following types of error hampered progress: ♦ incorrect solution to the auxiliary equation ♦ incorrect trial functions for the particular integral with kxe^{-x} and kx^2e^{-x} common ♦ lack of care when differentiating the general solution ♦ complementary function and final answer given in complex form. Multistep questions of this nature should be treated with extra care, since an early error can result in the loss of several marks by changing the level of difficulty of the follow-through work. Centres may wish to draw candidates' attention to the specification where it lists the (three) possible trial functions that candidates have to recall – the form of others is always given when required.	
			Total	11		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
24			$\frac{dy}{dx} - 3\frac{y}{x} = x^3 e^{2x}$ $I = \exp\left(\int -\frac{3}{x} dx\right) = e^{-3\ln x}$ $= x^{-3}$ $x^{-3} \frac{dy}{dx} - 3x^{-4} y = e^{2x}$ $\frac{d}{dx}(x^{-3} y) = e^{2x}$ $x^{-3} y = \frac{1}{2} e^{2x} + A$ $x = 1, y = 0 \Rightarrow A = -\frac{1}{2} e^2$ $y = \frac{1}{2} x^3 (e^{2x} - e^2)$	M1 M1 A1 M1 M1 A1 M1 A1	Divide by x Multiply and recognise derivative Integrate Use condition <u>Examiner's Comments</u> This was another standard question from which most candidates were able to gain high marks. Common errors which caused progress to be impossible were not recognising the need to initially divide throughout by x or neglecting the negative sign in the initial equation. Where candidates successfully heeded these two issues, they could usually work through to a general solution. Some of them, however, lost the final mark by neglecting to multiply their constant by the x^3 term when isolating y . Candidates could pick up the final method mark, even when they had been unsuccessful in finding the general solution, by demonstrating that they knew how to find the constant using the boundary condition.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	PI: $y = ax \cos 2x + bx \sin 2x$			
		i	$\frac{dy}{dx} = a \cos 2x - 2ax \sin 2x + b \sin 2x + 2bx \cos 2x$ $\frac{d^2y}{dx^2} = -4a \sin 2x - 4ax \cos 2x + 4b \cos 2x - 4bx \sin 2x$ in DE:	B1	For correct $\frac{dy}{dx}$ or better	
		i	$-4a \sin 2x - 4ax \cos 2x +$ $4b \cos 2x - 4bx \sin 2x$ $+4(ax \cos 2x + bx \sin 2x)$	M1	Differentiate twice and substitute	
		i	compare coefficients: $-4a = 1, 4b = 0$	M1		
		i	$\Rightarrow a = -\frac{1}{4}, b = 0$	A1		
		i	AE: $\lambda^2 + 4 = 0$ $\lambda = \pm 2i$	M1	For correct auxiliary equation and attempt to solve	
		i	CF: $A \cos 2x + B \sin 2x$	A1	oe form	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	GS: $y = \left(A - \frac{1}{4}x\right)\cos 2x + B\sin 2x$	A1ft	Must be real and contain 2 unknowns <u>Examiner's Comments</u> Although this question covered another standard technique, there was a significant minority of candidates who made little progress towards a solution. Some demonstrated a lack of knowledge of the standard form of solution resulting from complex roots to the auxiliary equation. Others took insufficient care when finding derivatives of the particular integral. As in previous sessions, there were some whose work resulted in "constants" which were functions of x and others who failed to see any problem with a final solution which contained complex coefficients.	
		ii	oscillations	B1	oe (accept sketch) dep consistent with 6(i)	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	unbounded	B1	oe (accept sketch) dep consistent with 6(i) If zero, then sc1 for recognition that $x \cos 2x$ term becomes dominant <u>Examiner's Comments</u> Marks were gained by some candidates for a recognition that the $x \cos 2x$ term would dominate as x tended to infinity, but only a few were able to fully describe the behaviour of this function. Good answers often used a diagram to flesh out a verbal explanation.	
		iii	If $k \neq 2$ then Pl $y = \alpha \cos kx + \beta \sin kx$	B1		
		iii	So bounded oscillations	B1	oe (accept sketch) <u>Examiner's Comments</u> This question was, in part, a way to assess candidates understanding of how to select the particular integral since this is the change that affects the shape of the graph as x tends to infinity. Candidates who could do part (ii) and who spotted this usually gained full marks.	
			Total	11		